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REVIEW

Anaerobic digestion process: technological aspects and recent developments

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Abstract

The technology of anaerobic digestion allows the use of biodegradable waste for energy production by breaking down organic matter through a series of biochemical reactions. Such process generates biogas (productivity of 0.45 Nm^3/KgSV), which can be used as energy source in industrial activities or as fuel for automotive vehicles. Anaerobic digestion is an economically viable and environmentally friendly process since it makes possible obtaining clean energy at a low cost and without generating greenhouse gases. Searching for clean energy sources has been the target of scientists worldwide, and this technology has excelled on the basis of efficiency in organic matter conversion into biogas (yield in the range of 0.7–2.0 kWh/m^3), considered energy carriers for the future. This paper gives an overview of the technology of anaerobic digestion of food waste, describing the metabolism and microorganisms involved in this process, as well as the operational factors that affect it such as temperature, pH, organic loading, moisture, *C*/*N* ratio, and co-digestion. The types of reactors that can be used, the methane production, and the most recent developments in this area are also presented and discussed.

Keywords Biogas · Energy · Organic residues · Methane · Hydrogen

Introduction

Interest in biomass conversion processes has increased in recent decades due to the need to replace non-renewable energy sources by more sustainable alternatives, while adding value to the wastes, and improving economic returns and/or viability of processes. Partial replacement of fossil resources to produce chemical compounds, energy, and fuel can be achieved with an efficient use of renewable biomass, such as agricultural waste and municipal solid waste (MSW) (Forster-Carneiro et al. 2013; Lima et al. 2011; Lucke 2012; Mamede 2013). Due to the huge amount available, biomass represents an attractive feedstock for the production of chemicals, fuels, and energy, being, therefore, a resource that inspires the bioeconomy.

Bioenergy, for example, is a renewable and clean energy derived from biomass and wastes, which has attracted great attention nowadays due to the concern about the depletion of fossil fuel reserves and an increase in greenhouse gases production. The energy from biomass can be produced through biochemical, biological, and thermochemical processes (Ni et al. 2006; Zhang et al. 2010). A variety of biomass sources and wastes can be used to produce energy, which can be classified as energy crops (herbaceous energy crops, woody energy crops, industrial crops, agricultural crops, and farms); agricultural wastes (crop residues and animal waste); forestry residues (trees and shrubs); and industrial and municipal waste (MSW, sewage sludge, industrial waste, and treatment of liquid phases of municipal waste) (Ni et al. 2006; Daija et al. 2016; Rikmann et al. 2017; Zekker et al. 2017).

Organic residues generated in industrial processes and from human activities have called the attention of authorities from several countries not only due to the large amount generated but also due to the impacts caused by their disposal

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(Lin et al. 2014). The amount of these wastes depends on several factors such as population growth, industrialization, and high consumption of products by the society. The inadequate management of organic residues has become a prominent issue for some countries including Brazil. Currently, the landfill is still the most common final destination for wastes in Brazil. However, the current legislation suggests targets for biodegradable waste elimination and aftercare in order to prevent or reduce negative effects on the environment and human health, that is, to reduce, reuse, or recycle solid wastes in order to recover materials and energy, conserve natural resources by avoiding environmental degradation, and especially achieve a better quality of life (European Parliament 2008; EPA 2013; BioCycle 2014; EEA 2014; EPE 214). In addition, in Brazil these residues have high organic loading (approx. 60% of all solid waste is organic) and, in many situations, they are transported, without reuse, to landfills or transfer stations located outside the urban area (EPE 2014). This aspect involves very significant logistics costs and is, therefore, an important variable in decision making regarding final disposal of MSW. The disposal of such wastes is a complex operation and leads to environmental impacts that go beyond the limits of the municipalities (EPE 2014). This problem with the waste disposal may be minimized by the use of biological reactors such as MBBR (moving bed biofilm reactor), SBR (a sequencing batch reactor), and UASB (upflow anaerobic sludge blanket) because some studies already prove the efficiency of these systems in the waste treatment with high organic loading (Raudkivi et al. 2017; Zekker et al. 2016).

The biorefinery concept, which defines and drives the use of biomass to produce materials, chemicals, energy, and heat by means of various technologies into a single industrial plant, can be applied to solid wastes. Furthermore, the biogas produced in an anaerobic digestion plant in a biorefinery could be burned to produce heat and electricity, while the produced electricity could be used for consumption or sold in dealer networks (Petruccioli et al. 2011). The biorefinery concept applied to biomass and solid wastes has significant environmental benefits, stimulates technological innovation to develop new products, maximizes recycling, extends landfills life cycle, and provides jobs, in addition to be an alternative to meet energy demand and conserve natural resources. Wastes of fruits and vegetables produced in large quantities in all markets around the world are able to degrade rapidly, constituting an ideal substrate for conversion technologies such as anaerobic digestion (AD). Some wastes of fruits and vegetables already used in methane production by anaerobic degradation presented productivity of 0.45 Nm^3/KgSV and feature a fairly biodegradable chemical composition mostly made up of carbohydrates, proteins, lipids, and minerals (Garcia-Peña et al. 2011; Girotto et al. 2015; Poggi-Varaldo et al. 2014). These elements are easily degraded by anaerobic bacteria and can be converted to biogas by AD, while the digested material can be further used as fertilizer on fields. Table 1 summarizes the physicochemical characteristics of different food wastes.

The treatment of waste and water with high organic loading may be performed by *biomethanation or* anaerobic co-digestion, which is an anaerobic process that occurs in reactors under mesophilic (35 °C) or thermophilic (55 °C) conditions, allowing the destruction of pathogens, degradation of high organic material, and production of biogas. Anaerobic processes of dry digestion (high percentage of

Table 1 Physicochemical characteristics of different food wastes

Food waste	Moisture (%, w/w)	TS (%)	VS (%)	*C*/*N*	References
Food waste from a university campus restaurant, Cádiz-Spain	10.10	89.80	69.80	35.40	Forster-Carneiro et al. (2008)
Food waste from canteens and restaurants of University of Cádiz, Puerto Real, Spain	68.9	31.1	27.2	37.4	Angeriz-Campoy et al. (2017)
Primary waste from university canteen	77.73	22.77	13.87	–	Gaur and Suthar (2017)
Food waste from dining room at Tongji University in Shanghai	73.5	26.5	25.09	13.4	Yi et al. (2014)
Synthetic food waste	76.30	23.70	21.60	–	Ariunbaatar et al. (2015)
Food waste from cafeteria of Quaid-i-Azam University, Islamabad, Pakistan	72.55	27.45	91.99	16.81	Jabeen (2015)
Food waste from a canteen of Beijing University of Chemical Technology (BUCT)	79.95	20.05	19.21	28.40	Yong et al. (2015)
Food waste from local residences at Ireland	74.2	25.8	20.3	–	Dennehy et al. (2017)
Food waste from restaurants, Kyonggi University, South Korea	76.98	23.02	20.55	–	Nguyen et al. (2017)

TS total solid; *VS* volatile solid; *C* carbon; *N* nitrogen

solids in the reactor) have high efficiency and can be used as a solution for solid waste. An important advantage of the anaerobic process is the possibility of producing hydrogen (H_2) in a stage known as acidogenic, and of producing methane (CH_4) in a stage known as methanogenic, being both products considered as energy vectors of the future (Cesaro et al. 2012). In this context, the technology of AD should be considered following biorefinery principles, being energy production from MSW the most important feature, as it allows economies of scale, that is, investment per unit of input decreases and conversion efficiencies increase with capacity (FEAM 2012). The present paper gives an overview on the importance of the AD technology. The most recent advances that have been reported for this technique, the metabolism and related microorganisms, the operational factors that affect this process including temperature, pH, organic loading, moisture, *C*/*N* ratio, and co-digestion, and the types of reactors that can be used are also presented and discussed.

Anaerobic digestion and biogas production

AD is a natural process that occurs in the absence of oxygen, and usually takes place in watercourses, sediments, and wet soils, from raw materials like industrial and municipal wastewater, agro-industrial, municipal, food activities, and vegetal wastes, resulting in the production of a gas mixture known as biogas (Kythreotou et al. 2014; Ward et al. 2008). AD is the only current biological treatment process able to produce biogas, which is a mixture of methane (CH_4) and carbon dioxide (CO_2). Instead of composting, AD allows biodegradation of organic matters in the absence of oxygen, resulting in a digested product that can be used as soil fertilizer. AD allows the treatment of high organic loading wastes in order to reduce its volume and load, while recovering biogas, which can be used to produce heat, electricity, and biofuels for vehicles (Awe et al. 2017).

The technology of AD can be classified according to the dry matter of the initial substrate as: wet digestion, where the substrate must have a dry matter content below 10%; dry digestion, in which the substrate should have dry matter content exceeding 20%; and dry matter digestion intermediate, also defined as semi-dry (Gkamarazi 2015). Wet AD was largely employed in the first anaerobic reactors for treatment of sewage sludge in wastewater treatment plants, as well as for treatment of liquid waste and/or organic waste with high moisture content and low organic loading. Dry AD was specifically developed to approach treatment of non-separated MSW or organic waste rich in dry fractions or badly diluted. Usually, the biogas produced in these processes is collected in the gas tank and it can be directly exported to the national gas system or sent for combustion in cogeneration system to generate electricity (with a yield in the range of 0.7–2.0 kWh/m^3 biogas) and heat (Cheng and Wang 2013).

AD process is a complex dynamic system involving microbiological, biochemical, and physical–chemical processes. Among the possible methods for treating biowaste, AD has been identified as the most environmentally sustainable option since it makes possible not only to deviate biodegradable materials from landfills, but also to produce bioenergy and potential by-products, such as soil biofertilizer (Monson et al., 2007). The project described by Gkamarazi (2015), for example, uses discontinuous reactors for dry AD of solid waste and was designed to work with a total throughput of 80.000 tons of MSW per year. According to the document on the best techniques available for treatment of industrial wastes (European-Commission, 2006), the AD process leads to a theoretical methane production of 348 Nm^3/ton of chemical oxygen demand (COD). This methane production depends on the raw materials, but typically, 80–200 m^3 CH_4 per ton of MSW, has been reported in the literature (Mes et al., 2003). Several studies in the literature have shown that the AD process is a robust process with efficient treatment system and high rates of organic load removal (Berni et al. 2014; Jabeen 2015; Mamimin et al. 2015).

Anaerobic digestion technology

With the energy crisis of 1970, AD process was taken into consideration, and it has been now increasingly used due to the emergence of immobilized biomass systems to treat sewage (Foresti et al. 2006). Organic matter can be converted, in the absence of oxygen, by different groups of specific microorganisms to methane (CH_4) and carbon dioxide (CO_2) by AD (Forster-Carneiro et al. 2008). Furthermore, from an environmental point of view, the anaerobic process reduces the content of organic matter present in the waste, making them less polluting when discarded into the environment and able to be disposed as fertilizer (Tambone et al. 2010). In relation to energy aspects, the anaerobic process enables the generation of biogas that can be used as an alternative source of energy. This is because the biogas generated by AD has a high heat of combustion (Uggetti et al. 2014). Table 2 shows AD advantages and disadvantages compared to the aerobic process. Biogas composition generated in the anaerobic

Table 2 Advantages and disadvantages of the anaerobic digestion process

Advantages	Disadvantages
• Lower production of biomass sludge compared to aerobic processes (Ward et al. 2008); 10% converted into mud and the other 90% into biogas (Khanal 2009)	• Long time for stabilization of the microbial culture within the reactor (Khanal 2009)
• A carbon neutral energy source is achieved in the form of biogas (Ward et al. 2008)	• Complex materials require pre-treatment (Braber 1995)
• The final residue after digestion is rich in nutrients and can be used in agriculture as fertilizer (Tambone et al. 2010)	• Post-treatment of the waste generated by the process before discharge to the environment can be necessary
• More cost-effective than other biological processes for energy recovery with low environmental impact (Kothari et al. 2014)	• Constant monitoring of key parameters such as pH, temperature, feed rate, and production of inhibitors is required (Khanal 2009)

process may vary with organic matter, microorganisms, and process conditions. According to Awe et al. 2017 in AD the organic matter is transformed by microorganisms action in CH_4 (55–70%), CO_2 (30–45%), and trace of elements such as H_2S, H_2, and N_2. Wukovits and Schnitzhofer (2009), on the other hand, reported that the proportion of gas in the biogas resulting from AD varies 50–75% CH_4, 25–45% CO_2, and 2–7% H_2.

Microbiology of AD

Generally, four trophic groups are considered relevant to AD processes, namely hydrolytic, acidogenic, acetogenic, and methanogenic bacteria (Kwietniewska and Tys 2014). Carbohydrates, lipids, and proteins are broken by hydrolytic bacteria into sugars, long-chain fatty acids, and amino acids, respectively. Then, these molecules are converted into volatile fatty acids, alcohols, CO_2, and H_2 in the acidogenic phase. These molecules are further converted by acetogenic

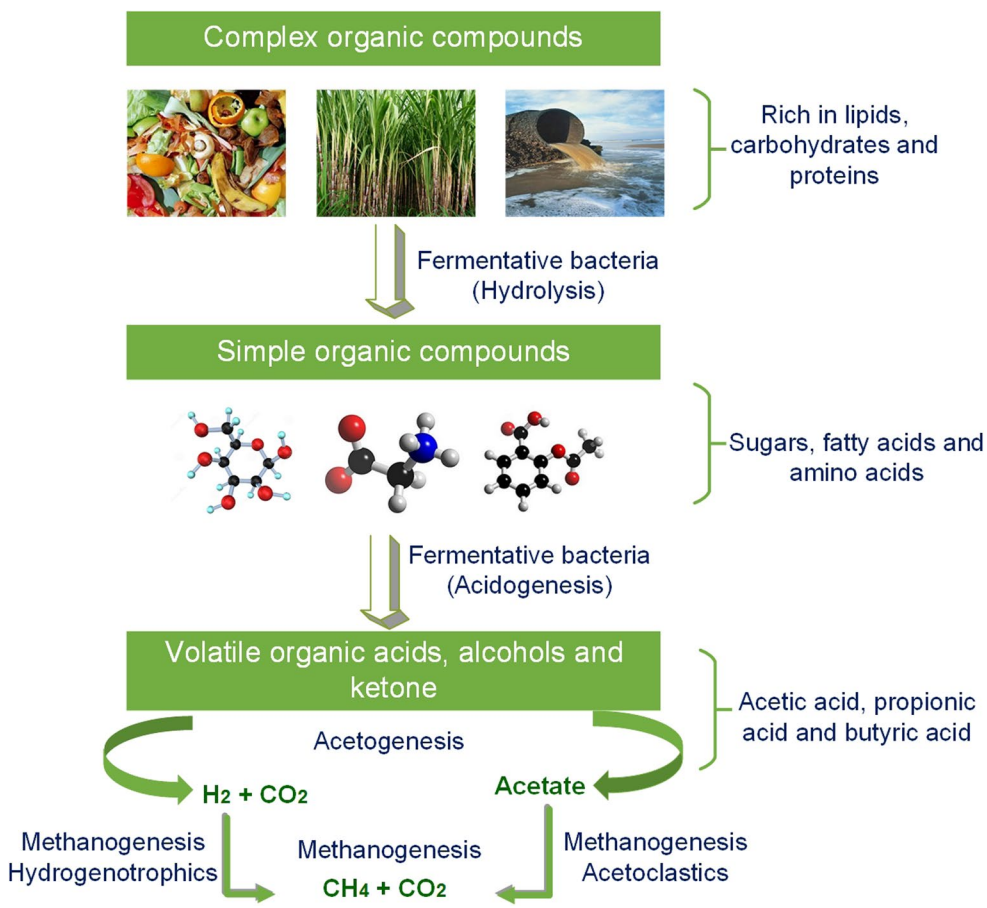

Fig. 1 Main steps of anaerobic digestion process

bacteria mainly to acetic acid, H_2, and CO_2. Finally, all these intermediate products are transformed into CH_2, CO_2, and water in the last stage, where methanogenic bacteria are involved. Figure 1 depicts the different steps of AD.

Hydrolytic microorganisms

During the first phase of the AD process, the most complex particulate organic matter (polymers) is transformed into dissolved simple materials. This conversion process is given by the activity of hydrolytic microorganisms (*Clostridia, Micrococci, Bacteroides, Butyrivibrio, Fusobacterium, Selenomonas, Streptococcus*) and requires the production of exoenzymes (cellulase, cellobiase, xylanase, amylase, protease, lipase) excreted by the fermentative bacteria, which break down complex compounds such as proteins, amino acids, and carbohydrates into mono- and disaccharides and also enable the conversion of lipids into long-chain fatty acids and glycerin (Chernicharo 1997; Merlin Christy et al. 2014). This step is the most time-consuming of the AD process due to the formation of toxic or unwanted compounds (Ariunbaatar et al. 2014). Hydrolysis also limits the AD speed if the used substrate is in the form of particles (Bouallagui et al. 2005). Then, an intensification of hydrolysis process leads to an increase in the performance of digestion (Yu et al. 2016). Biological, chemical, and mechanical pre-treatments, or a combination thereof, can be used to accelerate hydrolysis, because they can cause lysis or disintegration of the substrate and allow the release of intracellular matter allowing greater accessibility of anaerobic microorganisms, thus reducing the retention time in the digester (Ferrer et al. 2008).

Acidogenic microorganisms

The second phase of AD occurs due to the action of acidogenic fermentative microorganisms (*Streptococcus*, *Lactobacillus, Bacillus*, *Escherichia coli*, *Salmonella*), which convert soluble products of hydrolysis into compounds such as fatty acids, alcohols, lactic acid, carbon dioxide, hydrogen, ammonia, and hydrogen sulfide (Chernicharo 1997; Merlin Christy et al. 2014). The conversion of organic material into organic acids causes the pH of the system to drop. This acidic condition favors the action of acidogenic and acetogenic microorganisms, which prefer a slightly acidic pH (4.5–5.5). The acetic and butyric acids produced in this step are important in the overall performance of the AD, since these acids are the preferred precursors for methane production, which occur in the last step of the organic material degradation process (Hwang et al. 2001).

Acetogenic microorganisms

In the third phase, acetogenic bacteria convert the compounds generated during the acidogenic phase, producing hydrogen, carbon dioxide, and acetate. During acetic and propionic acids formation, a large amount of hydrogen ions is formed, causing a decrease in the pH of the aqueous medium (Chernicharo 1997; Mes et al. 2003; Gkamarazi 2015). The optimum pH for the action of acetogenic microorganisms is around 6. These microorganisms are slow growing and sensitive to fluctuations in organic loadings and environmental changes (Merlin Christy et al. 2014).

Methanogenic microorganisms

The last stage of the AD process is called methanogenesis. In this phase, methanogenic archaea promote the breakdown of organic compounds derived from the acetogenic phase. Methanogenic archaea are divided into two main groups: acetoclastic, which degrades acetic acid or methanol to produce methane, and hydrogenotrophic, which uses hydrogen and carbon dioxide to produce methane (Chernicharo 1997; Mes et al. 2003; Gkamarazi 2015). Hydrogenotrophic methanogens are more resistant to environmental changes than acetoclastic methanogens. Unlike other microorganisms, methanogens prefer slightly alkaline environment (Merlin Christy et al. 2014) around 6.5–8 (Kothari et al. 2014). A well-balanced AD process occurs when all the products obtained in a metabolic stage are converted in the next stage without accumulation of intermediate products, resulting in a complete breakdown of organic matter in final products of interest, such as methane, carbon dioxide, hydrogen sulfide, and ammonia (Bouallagui et al. 2005). However, if the objective is to ensure hydrogen production by means of fermentative processes, it is a essential blockage or inhibition of methanogenic bacteria by controlling the temperature, pH, or even by using additives, since such bacteria consume hydrogen for methane production (Wang and Zhao 2009).

Effect of operational factors on AD

Temperature

Temperature causes significant effects on microbial communities, interfering in the stability of the process, microbial growth, substrate utilization rate, and biogas production (Khalid et al. 2011). Four temperature schemes can be used in anaerobic digesters for biogas formation (Forster-Carneiro et al. 2013; Kothari et al. 2014; Kwietniewska and Tys 2014): (1) psychrophilic temperatures: 10–20 °C with

optimum at 25 °C; (2) mesophilic temperatures: 20–45 °C with optimum at 35 °C; (3) thermophilic temperatures: 50–65 °C with optimum at 55 °C; and (4) extremely thermophilic temperatures: between 65 and 70 °C.

Under psychotropic conditions, chemical and biological reactions occur more slowly (Trzcinski 2009), while under thermophilic conditions, the metabolic rate of microorganisms increases. Therefore, thermophilic temperatures are commonly used in large-scale biodigesters (Kothari et al. 2014; Kwietniewska and Tys 2014). In addition, thermophilic temperatures ensure higher rates of pathogens destruction (Ho et al. 2014). However, the use of high temperatures requires higher energy costs, an increased process control to achieve uniform, and constant temperature inside the biodigester (Kwietniewska and Tys 2014) and thermophilic biodegradation may favor the acidification of the reactor by inhibiting biogas production (Mao et al. 2015).

Mesophilic temperatures are the most popular used in anaerobic digesters for biogas formation, since they allow obtaining good digestion with lower energy costs (Kothari et al. 2014). Moreover, mesophilic temperatures provide higher stability of the process and lower sensitivity to environmental changes (Mao et al. 2015). On the other hand, mesophilic microorganisms are more affected by ammonia accumulation than the thermophilic ones (Wang et al. 2016). However, a change from mesophilic to thermophilic temperatures can cause a decrease in biogas production until the microbial communities adjust to the medium, and then there is an increase in the number of microbial communities (Khalid et al. 2011; Ward et al. 2008).

According to Briški et al. (2007), the breakdown of organic matter must occur at temperatures lower than 65 °C, as higher values promote enzyme's denaturation. Hegde and Pullammanappallil (2007) observed that the breakdown of fatty acids at 55 °C was faster than at 38 °C. The retention time was also 95% shorter when the process was performed at 38 °C. A comparison of mesophilic and thermophilic conditions was also studied by Yu et al. (2014). According to these authors, for an organic loading of 3 kgVS m^{-3} d^{-1} at 55–57 °C, the biogas production was 749.69 l $kgVS^{-1}$, whereas at 35–37 °C the production was 628.80 l $kgVS^{-1}$. Nevertheless, as the organic loading increased, greater biogas production was observed at mesophilic temperatures than at thermophilic temperatures, in which, to an organic loading of 8 kgVS m^{-3} d^{-1}, the production achieved 536 l $kgVS^{-1}$ under thermophilic conditions and 628 l $kgVS^{-1}$ under mesophilic conditions.

pH

Each microorganism grows better at a specific pH range, and the maximum microbial growth occurs at an optimum pH value (Montañés et al. 2014). Anaerobic bacteria involved in the AD process require different optimum pHs for their growth (Azman et al. 2015). However, the ideal pH range for AD is very narrow, usually ranging between 6.8 and 7.2 (Ward et al. 2008), varying according to the substrate used in the process and the digestion technique employed (Zhai et al. 2015). The growth rate of methanogenic bacteria is reduced in environments with pH below 6.6, while a very alkaline pH (around 8) may lead to disintegration of microbial granules and subsequent failure of the process. In addition, regarding high alkalinity issues on most cases, the biodegradation process produces bicarbonate alkalinity, which neutralizes the acidity that is produced from the biodegradation process (Tenno et al. 2016). Therefore, an ideal pH for methanogenic phase is usually around 7 (Ward et al. 2008).

Acetogenic and methanogenic bacteria are more difficult to adapt to pH changes than other microorganisms. That is because acetogenic bacteria produce volatile organic acids that are converted into acetic acid, hydrogen, and carbon dioxide, which results in acidification of the environment that negatively influence the activity of methanogenic bacteria (Montañés et al. 2014). The optimum pH for H_2 production is 5 (Stavropoulos et al. 2016), while lower values (pH around 4) favor the production of acetic and butyric acids, and higher values (pH 8.0) favor the production of acetic and propionic acids (Appels et al. 2008).

Retention time

Hydraulic retention time (HRT) can be defined as the time required for a complete breakdown of organic matter or the time in which the organic material remains in the biodigester. HRT depends on the temperature of the process and substrate composition to be digested and may be defined by Eq. 1, where HRT is the hydraulic retention time (day), V is the volume (m^3), and Q is the daily flow (m^3/day) (Kothari et al. 2014).

$$\mathrm{HRT} = \frac{V}{Q} \tag{1}$$

Usually, the HRT for mesophilic microorganisms ranges from 10 to 40 days, while for thermophilic microorganisms the time is shorter, 14 days (Kothari et al., 2014). Hydrogen-producing bacteria prefer short retention times since they tend to produce volatile fatty acids and hydrogen in the exponential phase and alcohols in the stationary phase. As methanogenic bacteria consume hydrogen to produce methane and carbon dioxide, when they are inhibited, a higher H_2 yield is obtained.

In contrast, there is a decrease in methane production by methanogenic bacteria in short retention times. Methane production from AD of wheat straw, for example, was increased by increasing the retention time from 20 to 60 days; in addition, the digestion with HRT of 20 days showed lower stability compared with HRT of 40 and 60 days (Shi et al. 2017).

Organic loading

Organic loading rate (OLR) is the content of volatile solids that daily feeds the reactor per cubic meter of digester volume. OLR can be estimated through Eq. 2, where OLR is the organic loading rate [kgVS $(m^3 day)^{-1}$], V is the volume [m^3], Q is the daily flow [kg/day], and VS is the volatile solids [kgVS$(kg)^{-1}$] (Kothari et al. 2014):

$$\mathrm{OLR} = \frac{Q * \mathrm{VS}}{V} \tag{2}$$

Organic loading in the reactor is a very important control parameter in continuous systems (Kothari et al. 2014). Increasing the feeding rate causes higher biogas yield. However, if the feeding is very intense, an overload in the reactor occurs due to the accumulation of inhibitory substances such as volatile fatty acids (VFA). In the presence of high feeding volumes, bacteria responsible for hydrolysis and acidogenesis phases produce a high amount of VFA in a short time, leading to an acidification of the medium, inhibiting hydrolysis process, and preventing the performance of methanogenic bacteria that will not be able to convert substrates produced by the earlier stages in methane. A decrease in biogas production then occurs, consequently (Kwietniewska and Tys 2014; Mao et al. 2015).

AD can be classified into three categories according to the solids content used to feed the system (Kothari et al. 2014): low solids content (solids content less than 15%), medium solids content (solids content ranging between 15 and 20%), and high solids content (solids content between 20 and 40%).

Moisture

Water is an essential component of the organic matter breakdown process since it acts as solvent and contributes to mass transfer and diffusion of microorganisms, allowing interaction between the surface of the substrate with microorganisms involved in the process (Bollon et al. 2011; Vavilin et al. 2003). According to the moisture content, the AD processes can be classified as wet or dry. Dry processes operate with 30–40% dry matter, while wet systems operate with 10–25% dry matter (Kwietniewska and Tys 2014).

Biogas production from organic breakdown requires aqueous environments with water activity higher than 0.91. According to Kwietniewska and Tys (2014), the highest methane production occurs at 60–80% moisture since high levels of moisture facilitate digestion process (Khalid et al. 2011).

C/N ratio

C/N ratio is the ratio between the amount of carbon and nitrogen contained in the organic matter. This ratio plays an important role in the AD processes as the AD rate is affected by the type, availability, and complexity of the substrate (Khalid et al. 2011; Kothari et al. 2014). Carbohydrates present in MSW are organic compounds of great importance for biogas production (Khalid et al. 2011). Addition of carbohydrates increases protein conversion and activity of proteases in the sludge (Mao et al. 2015). Nitrogen is essential for protein synthesis and is used as a nutrient by the microorganisms responsible for AD. Nitrogen compounds from organic waste are converted into ammonia in the anaerobic degradation process. In the form of ammonia, nitrogen contributes to maintaining the pH of the medium stable during the process (Khalid et al. 2011). Thus, insufficient amounts of carbon and nitrogen can compromise the entire process of AD.

High *C/N* ratio means that a high carbon content is available, while low *C/N* ratio means that the material is rich in protein (Kwietniewska and Tys 2014). If this ratio is high, the methanogenesis quickly consumes the nitrogen, resulting in a low gas yield, but if the ratio is low, an accumulation of ammonia occurs. When the pH of the medium is greater than 8.5, the activity of methanogenic bacteria is negatively influenced (Kothari et al. 2014). According to Kwietniewska and Tys (2014), *C/N* ratio for AD of organic waste should be between 20 and 25. Yong et al. (2015) obtained maximum biogas production using *C/N* ratio of 31, and Dadaser-Celik et al. (2016) found an optimum *C/N* ratio for biogas production of 28.

Anaerobic co-digestion

Anaerobic co-digestion is a technology where two substrates are anaerobically digested for biogas production. It is usually employed to improve the efficiency of AD due to the following benefits: dilution of toxic compounds, increase in biodegradable organic matter, economic advantages due to equipment sharing, decreased nutrient imbalance adjusting N and C contents, higher biogas efficiency, and synergic

effect of microorganisms present in the reactions facilitating the use of mixed substrates (Kwietniewska and Tys 2014).

Co-digestion is a technology that allows optimizing the *C*/*N* ratio of substrates more easily in order to achieve higher biogas production (Abouelenien et al. 2016). Co-digestion provides greater stability in the reactor feeding, improving the *C*/*N* ratio by decreasing the nitrogen concentration and reducing problems with the accumulation of volatile compounds and ammonia (Khalid et al. 2011). In addition, combining substrates in the co-digestion process makes possible to meet the nutritional needs of the microbial community, favoring their activities and promoting higher biogas yield. According to Sialve et al. (2009), the needs of anaerobic microorganisms in terms of carbon and nitrogen are not met when the *C*/*N* ratio is less than 20.

Several studies have demonstrated the efficiency of using co-digestion processes for biogas production. Ye et al. (2013), for example, used 2.5-L reactors and mesophilic conditions to compare mono-digestion with co-digestion of rice straw with kitchen and manure wastes from pigs and observed a 71% increase in methane production when using the co-digestion technology. The proportion of wastes used in their process was 0.4:1.6:1 for kitchen waste, pig manure, and rice straw, respectively. Aichinger et al. (2015) used two 10-L continuously stirred mesophilic laboratory scale digesters and also observed an increase of about 35% in biogas production when organic fraction municipal waste was co-digested with thermally pre-treated primary and waste-activated sewage sludge. Santos et al. (2014) studied the production of hydrogen at thermophilic temperatures by co-digestion of cane vinasse and glucose and achieved a maximum H_2 yield of 5.73 mmol/g COD_{added}, in which hydrogen represented 50.8% of biogas.

Regarding pollutants removal for biogas effluent wastewater for fixed bed and moving biofilm reactors, the biological treatment MBBR systems could be efficient being dependent on temperature and toxic nitrite concentrations (Zekker et al. 2013, 2012). Thus, anaerobic ammonium oxidation (anammox) process has been put forward as a novel and promising biological nitrogen removal process by researchers in the field of wastewater treatment, and a number of investigations have been made on biological treatment of N-rich wastewaters (Rikmann et al. 2014).

Anaerobic reactors

The reactors used for AD must be able to allow a continuous high organic load rate, low hydraulic retention time, and should have also capacity for a high biogas production (Ward et al. 2008). The reactors that have been used for AD can be classified as single-phase reactors, multiphase reactors, batch reactors, and reactors that operate in continuous or semi-continuous process (Kothari et al. 2014).

In single-phase processes, a single reactor is used for the acidogenic and methanogenic phases, while multiphase processes require more than one reactor (generally two) to separate the main process stages (acidogenic and methanogenic). A two-phase system is considered promising for organic waste treatment because it improves the conversion of organic matter, since optimum conditions can be applied in each reactor, thereby increasing process efficiency (Khalid et al. 2011). In this kind of system, the first phase aims to optimize the hydrolysis and acidogenesis reactions, whose rate is limited by hydrolysis of complex carbohydrates (Kothari et al. 2014). This stage occurs at low pH as the acidogenic bacteria convert complex organic matters into volatile fatty acids and alcohols. The second phase aims to optimize the methanogenesis reactions, by converting the products obtained in the previous phase into carbon dioxide and methane. The action of methanogenic bacteria occurs preferably at pH between 7 and 8.5 (Khalid et al. 2011; Kothari et al. 2014). Multiphase reactors are more expensive to build and maintain than single-phase reactors, but they are the most used as they present a better performance (Ward et al. 2008).

In terms of process operation, batch reactors are the simplest models. Such reactors are fed with a certain

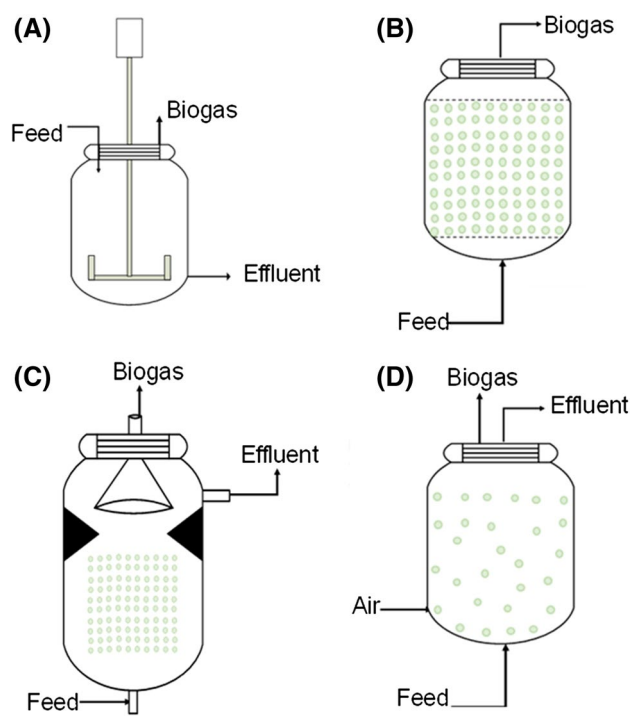

Fig. 2 Anaerobic reactors: **a** CSTR (continuous stirred-tank reactor); **b** fixed bed reactor; **c** UASB (upflow anaerobic sludge blanket); **d** fluidized bed reactor

amount of organic matter, which is kept inside the reactor for a period (which can be considered as HRT), being the reactor emptied after completed the reaction (Ward et al. 2008). Anaerobic batch reactors are useful because digestion can occur quickly, with simple equipment and low cost. However, these reactors have limitations such as high fluctuations in gas production, biogas quality, and biogas losses when emptying the reactors (Khalid et al. 2011). In the continuous scheme, on the other hand, the reactors are continuously fed with organic matter and the same amount of digested matter is continuously removed from the reactor. In continuous processes, chemical reactions occur at a relatively constant rate resulting in constant biogas production. The main disadvantage of this process is that partially digested matter is also removed from the reactor during the effluent removal (Kothari et al. 2014).

The basic features of an anaerobic reactor are allowing high organic rate and short HRT in order to reduce the reactor volume and enable high biogas productivity (Ward et al. 2008). Various types of reactors can be used for biogas production, among which the continuous stirred-tank reactor (CSTR), fixed bed reactor, upflow anaerobic sludge blanket (UASB) reactor, and fluidized bed reactor (Fig. 2) are the most commonly employed.

CSTR

CSTRs are simple reactors that were adapted from pond-type anaerobic reactors. In such reactors, an agitator allows the contact between microorganisms and effluent, avoiding sedimentation of solids within the reactor and diminishing the resistance to mass transfer, which allows the release of trapped gas bubbles in the medium (Mao et al. 2015; Ward et al. 2008). Agitation may occur continuously or intermittently, being driven in a time interval (Ward et al. 2008). The disadvantages of this type of reactor include the restriction on biogas production, since this reactor model does not maintain high levels of biomass due to constant stirring (Show et al. 2011) and the energy expenditure to maintain stirring. A limiting factor for hydrogen production using a CSTR is the stirring speed (Arimi et al. 2015). According to Ri et al. (2017) the optimum stirring speed for hydrogen production is 50 rpm. Excess of stirring is detrimental to biogas production, since microbial granules suffer disruption in their structure and reduce fatty acids oxidation. Thus, they accumulate inside the reactor acidifying the medium and causing inhibition of AD process (Mao et al. 2015; Ward et al. 2008).

Fixed bed reactor

Fixed bed reactors have in their interior a microbial support material that can be inert or biodegradable, made from different materials including activated carbon, ceramics, among others (Show et al. 2011; Ward et al. 2008). The porosity of the material used for biomass immobilization determines the fluid dynamics in the reactor (Arimi et al. 2015; Show et al. 2011). Fixed bed reactors are characterized by a low hydraulic turbulence, which causes mass transfer difficulties for the immobilized cells, resulting in low substrate conversion and low hydrogen production (Kothari et al. 2014). Such behavior occurs due to the low turbulence, which makes difficult to maintain a homogeneous pH, causing a heterogeneous distribution in the microbial activity (Show et al. 2011). Recirculating the biomass is an alternative to increase the production rate and product yield in such type of reactors, since it reduces the mass transfer resistance (Arimi et al. 2015; Show et al. 2011). The main disadvantages of this system are the cost of investment and energy expenditure (Arimi et al. 2015).

UASB reactor

The UASB reactor is considered one of the most versatile reactors for use in AD, since it retains a high amount of biomass and keeps constant the biomass to substrate ratio (Arimi et al. 2015). It is a simple technology, compact, and inexpensive and widely used in wastewater treatment. UASB reactors have at their bottom a dense sludge bed that favors the contact of wastewater with biomass. Some important advantages of this system include the stability and the fact that biogas production does not diminish with time. As disadvantages, it can be mentioned the long starting period, high retention time of solids, and high content of microorganisms in the wastewater (Arimi et al. 2015; Mao et al. 2015). The UASB reactors were developed for the effective domestic wastewater treatment (Rizvi et al. 2015). In recent years, e.g., the use of anammox UASB has been shown to be efficient in the removal of biological nitrogen during wastewater treatment (Rikmann et al. 2014) and has been shown to be an attractive way to remove wastewater pollutants like pharmaceutically active compounds (Alvarino et al. 2016).

Table 3 Reactors used for biogas production by anaerobic digestion and co-digestion

Substrate	Technology	Inoculum	Temperature (°C)	Reactor	HRT	Maximum yield	References
Food waste-water	Semi-continuous and single-phase co-digestion	Sludge from a municipal wastewater treatment plant	55	CSTR (5L)	20 d	CH_4: 316.11 mL (g $COD_{removed})^{-1}$	(Jang et al. 2015)
Chinese cabbage waste	Semi-continuous two-stage anaerobic digestion	Pig manure and sludge from wastewater treatment	37	CSTR (100 L)	12–16 d	CH_4: 0.62 $m^3(KgVS)^{-1}$	(Dong et al. 2015)
Food waste	Two-stage anaerobic digestion	Sludge from food waste treatment	55	H_2: CSTR (8L) CH_4: reactor of three chambers with agitation (40L)	H_2: 2.87 d CH_4:14.4 d	H_2: 147.3 L$(gVS_{add})^{-1}$ CH_4: 470 L$(gVS_{add})^{-1}$	(Kobayashi et al. 2012)
Palm oil mill effluent	Continuous two-stage anaerobic digestion	Sludge from palm oil mill effluent treatment	H_2: 55 CH_4: 35	H_2: ASBR 200 mL CH_4: UASB (3L)	H_2: 2 d CH_4:15 d	H_2 :210 mL$(gCOD)^{-1}$ CH_4: 315 mL$(gCOD)^{-1}$	(Mamimin et al. 2015)
Synthetic food waste	Continuous two-stage system anaerobic digestion	Sludge from treatment of buffalo manure and dairy wastes	H_2: 55 CH_4: 37	H_2: CSTR (1.6 L) CH_4: Fixed bed (1.3 L)	H_2: 3.5 d CH_4:1.5 d	H_2: 391.7 L H_2/m^3 CH_4: 2021.6 L CH_4/m^3	(Yeshanew et al. 2016b)
Synthetic carbohydrate-rich wastewater	Semi-continuous single-phase anaerobic digestion	Sludge from treatment of buffalo manure and dairy wastes	37	Fluidized bed 1.3 (L)	20 d	CH_4: 320 mL$(gCOD)^{-1}$	(Yeshanew et al. 2016a)

HRT hydraulic retention time, *CSTR* continuous stirred-tank reactor, *ASBR* anaerobic sequential batch reactor

Fluidized bed reactor

In this reactor, small particles, such as sand or alumina, are used to keep the microorganisms in suspension, ensuring their growth and a rapid upward flow of wastewater (Mao et al. 2015). This configuration allows a high solid retention time (Mao et al. 2015), good mixing within the reactor (Arimi et al. 2015), and good mass transfer between the microbial biomass and the wastewater. Low investment costs, due to the reduced volume of the fluidized bed reactors, make also them very attractive. This type of reactor is effective for the treatment of soluble material or when the feed material is suspended and readily biodegradable (Mao et al. 2015). Table 3 summarizes some studies using different reactors for biogas production from AD and co-digestion.

Conclusion

The technology of anaerobic digestion allows the use of biodegradable residues for energy production by breaking down the organic matter through a series of biochemical reactions. This technology is a promising process to produce energy as it makes possible obtaining high yields of H_2 and CH_4 that can be used as fuels in combustion engines, allows obtaining clean energy, reducing greenhouse gas emissions. A suitable control of the process parameters is of extreme importance for the success of anaerobic digestion process, since temperature, pH, HRT, OLR, moisture, and *C/N* ratio directly influence the behavior of the microorganisms responsible for biogas production. In terms of perspectives, anaerobic digestion has an enormous potential to contribute

to the management of organic residues of cities and industries, contributing, at the same time, to the diversification of the countries' energy grid, fostering research, generating knowledge and innovation on bioenergy and production of new products.

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